Day 28: Threat Intelligence / IOC Enrichment

Enriching the EICAR Test File Hash with VirusTotal and MISP Context

SOC Internship - TriosCyber

Field	Details
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Internship Track	SOC (Security Operations Center)
Assigned Task	Take a safe, non-sensitive IOC from instructor-provided sample data or a known training indicator. Check available reputation/context sources and record what additional context would affect analyst prioritization. (Tools: VirusTotal demo/sample IOC resources, Wazuh)
Status of this Report	**Practical exercise, fully completed with live lookups on 30 September 2026. The IOC is the SHA-256 hash of the EICAR antivirus test file, an industry-standard harmless training indicator. It was enriched from two independent sources, VirusTotal and the MISP EICAR warninglist, using hash search only: no file was downloaded, created or uploaded. The Wazuh portion describes Wazuh's documented VirusTotal integration; it was not run live, because no Wazuh manager is deployed in this lab (see Days 17-20).**

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Key Terms

- **IOC (Indicator of Compromise):** A piece of observable data that may point to malicious activity, such as a file hash, IP address, domain or URL.
- **Enrichment:** Looking an IOC up in external and internal sources to add context: what it is, how widely it is known, how old it is, and whether it is actually dangerous.
- **File Hash:** A fixed-length fingerprint of a file's exact contents (MD5, SHA-1, SHA-256). Identical files always produce identical hashes, so a hash can be searched without handling the file itself.
- **Detection Ratio:** The number of antivirus engines that flagged a file out of the number that returned a result, for example 66/68 on VirusTotal.
- **EICAR Test File:** A harmless text string created by EICAR (the European Institute for Computer Antivirus Research) and CARO (the Computer Antivirus Research Organization) that antivirus products detect as if it were malware, so that people can safely test whether their antivirus works. It contains no malicious code. The standard version is 68 bytes; the specification allows the file to be up to 128 bytes with trailing whitespace.
- **MISP Warninglist:** A curated list, published by the MISP threat-sharing project, of indicators known to be benign or non-actionable (such as test files and well-known public services). Its purpose is to stop analysts wasting time on false positives.
- **Prioritization:** Deciding how urgently an alert needs attention. The same indicator can be low priority in one context and critical in another.

Important Note on This Report

The indicator examined is the EICAR test file hash, which is harmless by design. Only its hash was searched; no file was downloaded, created or uploaded, and nothing was submitted to VirusTotal. The value of this exercise is not discovering that EICAR is benign, which is widely known, but demonstrating how raw reputation data can look alarming until context is added, and recording exactly which context changes an analyst's priority.

Executive Summary

The SHA-256 hash of the EICAR test file was enriched using two independent sources. VirusTotal reported that 66 of 68 security engines flagged it, with a popular threat label of virus.eicar/test and threat categories of virus and trojan. On detection count alone, this looks like a high-priority malware finding.

Context reverses that reading. Most vendor labels explicitly say the file is a test (for example "EICAR-Test-File (not A Virus)" and "NotAThreat.EICAR[TestFile]"), VirusTotal's file-type tools identify it as an EICAR antivirus test file, and the MISP project lists the same hash on its EICAR warninglist of known, non-actionable indicators. The indicator is therefore low priority by default.

The report records which additional context would change that default, most importantly where the file appeared, whether a test was scheduled, and whether EICAR is being used to probe or evade defenses. It also describes how Wazuh's VirusTotal integration would perform this enrichment automatically once a manager is deployed.

1. Objective

The objective of this lab is to take a safe, non-sensitive indicator of compromise, enrich it using publicly available reputation and context sources, and record which additional context would raise or lower its priority for a SOC analyst. The focus is not simply on whether the indicator is "bad", but on how an analyst turns raw reputation data into a justified triage decision.

To do this, the lab takes the SHA-256 hash of the EICAR antivirus test file as a known training indicator, because it is universally recognised by security tools yet completely harmless, so it can be investigated without any risk. The hash is searched on VirusTotal, the most widely used public multi-engine reputation service, to see how security vendors classify it, when it was first seen, and what VirusTotal's file-identification tools report. It is then checked against the MISP EICAR warninglist, an independent, community-maintained source, to confirm whether the indicator is known to be non-actionable. Using two sources matters because a single source can be misleading: a high detection count on its own looks alarming, and only corroborating context reveals what the indicator really is.

Finally, the lab records the environmental context that would change the indicator's priority, such as where it appeared, how it arrived and whether defenses stopped it, and describes how Wazuh's VirusTotal integration would automate the lookup step in a live deployment. Together, these steps reflect how enrichment is performed during real alert triage.

2. The Indicator

The EICAR test file was chosen because it is the industry-standard safe training indicator: every major antivirus product detects it, yet it contains no malicious code. Its hashes were calculated from the standard EICAR string and confirmed against VirusTotal (Figure 6) and the MISP warninglist (Figure 8):

Hash type	Value
SHA-256 (searched)	275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f
SHA-1	3395856ce81f2b7382dee72602f798b642f14140
MD5	44d88612fea8a8f36de82e1278abb02f

SHA-256 was used for the lookup because MD5 and SHA-1 are older algorithms for which deliberate collisions are possible; SHA-256 is the standard choice for identifying a file.

3. Source 1: VirusTotal

3.1 Searching Safely

Step 1: Search the hash, not the file



Figure 1. The VirusTotal home page with the SEARCH tab selected and the EICAR SHA-256 entered.

What this screenshot shows:

- VirusTotal offers three tabs: FILE uploads a file for scanning, URL submits a web address, and SEARCH looks up something VirusTotal has already seen.
- The SEARCH tab is selected and only the 64-character SHA-256 hash has been entered. This retrieves VirusTotal's existing record without sending any file.
- This is the safe habit to keep in a real SOC: files uploaded to VirusTotal are shared with the participating security vendors, and VirusTotal's premium subscribers can access files uploaded by others. Sensitive or internal files should therefore never be uploaded. Searching by hash only retrieves VirusTotal's existing record and shares no file.

3.2 Detection Summary

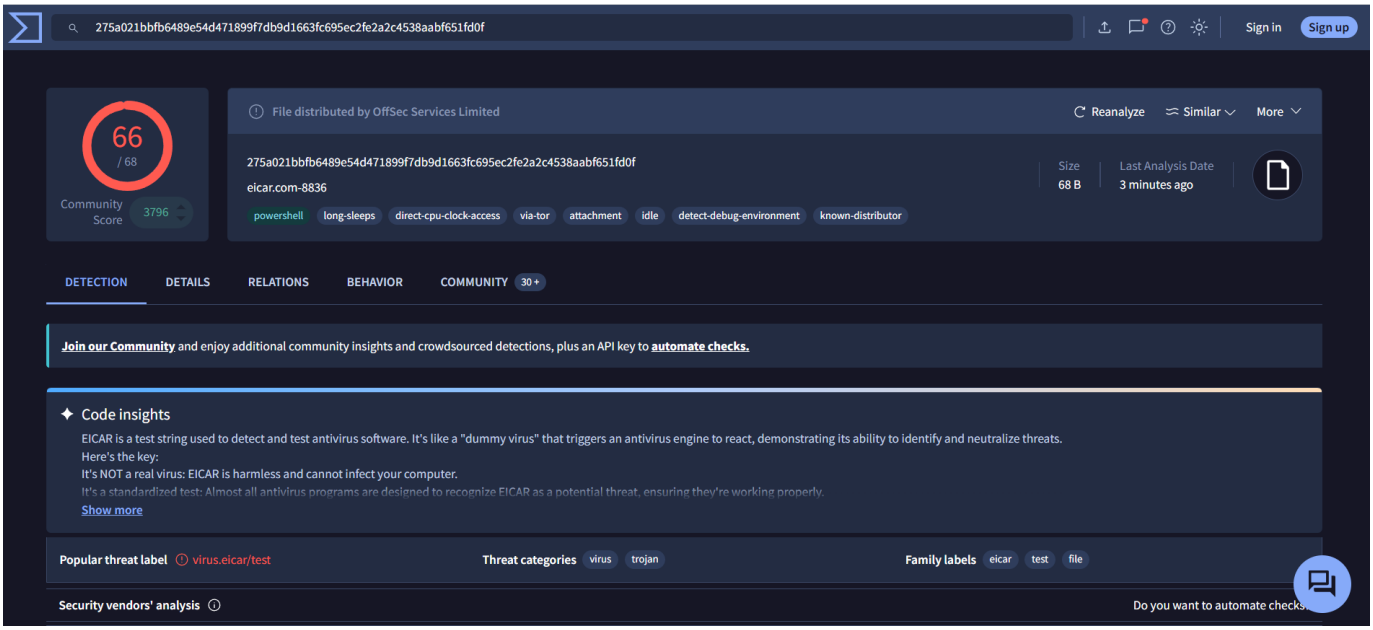


Figure 2. The VirusTotal result page for the EICAR hash, Detection tab.

What this screenshot shows:

- The red circle shows 66 / 68: 66 of the 68 engines that returned a result flagged the file. The Community Score of 3796 reflects votes from VirusTotal users.
- The file is 68 B, matching the size of the standard EICAR string, and the Last Analysis Date is "3 minutes ago", so the results reflect current engine signatures.
- The banner "File distributed by OffSec Services Limited" and the tag known-distributor indicate VirusTotal associates this file with a known, legitimate distributor, which is context against treating it as unknown malware.
- The Code insights panel is an automatically generated summary. It states that EICAR is a test string used to test antivirus software, that it is not a real virus, and that it is harmless and cannot infect a computer. As machine-generated text it is useful context, but the analyst still confirms it against other sources (Sections 3.4 and 4).
- Popular threat label virus.eicar/test, threat categories virus and trojan, and family labels eicar, test, file summarise how vendors classify it. The categories "virus" and "trojan" are how the detections are filed, not evidence of malicious behavior.
- The tags (powershell, long-sleeps, via-tor, detect-debug-environment and others) do not fit a 68-byte text string, and this page does not show where they came from. An analyst should not take automatically generated tags at face value.

3.3 Individual Vendor Verdicts



Popular threat label	virus.eicar/test	Threat categories	virus, trojan	Family labels	eicar, test, file
Security vendors' analysis				Do you want to automate checks?	
AhnLab-V3	Virus/EICAR_Test_File	Alibaba	Virus:Win32/EICAR.A		
AliCloud	Engtest:Multi/Eicar	ALYac	Misc.Eicar-Test-File		
Antiy-AVL	TestFile/Win32.EICAR	Arcabit	EICAR-Test-File (not A Virus)		
Avast	EICAR Test-NOT Virus!!!	Avast-Mobile	Eicar		
AVG	EICAR Test-NOT Virus!!!	Avira (no cloud)	Eicar-Test-Signature		
BitDefender	EICAR-Test-File (not A Virus)	Bkav Pro	W32.EicarTest.Trojan		
ClamAV	Eicar-Test-Signature	CMC	Eicar.test.file		
CTX	Txt.virus.eicar	Cynet	Malicious (score: 99)		
DrWeb	EICAR Test File (NOT A Virus!)	Elastic	Eicar		
Emsisoft	EICAR-Test-File (A)	eScan	EICAR-Test-File		
ESET-NOD32	Eicar Test File	Fortinet	EICAR_TEST_FILE		
GData	EICAR_TEST_FILE	Google	Detected		

Figure 3. Security vendors' analysis, first section.

What this screenshot shows:

- Each row is one security vendor's verdict. The red text is the name that vendor gives the file.
- Many names explicitly declare it a test: Arcabit and BitDefender "EICAR-Test-File (not A Virus)", Avast and AVG "EICAR Test-NOT Virus!!!", DrWeb "EICAR Test File (NOT A Virus!)".
- A few names sound more alarming without context, such as Bkav Pro "W32.EicarTest.Trojan" and Cynet "Malicious (score: 99)". This is why an analyst reads the labels rather than stopping at the score.



Figure 4. Security vendors' analysis, second section.

What this screenshot shows:

- Major vendors continue the same pattern: Kaspersky "EICAR-Test-File", Microsoft "Virus:DOS/EICAR_Test_File", Sophos "EICAR-AV-Test", Symantec "EICAR Test String".
- SUPERAntiSpyware labels it "NotAThreat.EICAR[TestFile]" and Sangfor Engine Zero "EICAR-Test-File (not A Virus)", explicitly stating it is not a threat.
- SentinelOne's machine-learning engine reports "Static AI - Malicious COM", a generic machine-learning verdict that does not name EICAR. Generic verdicts need the most context.

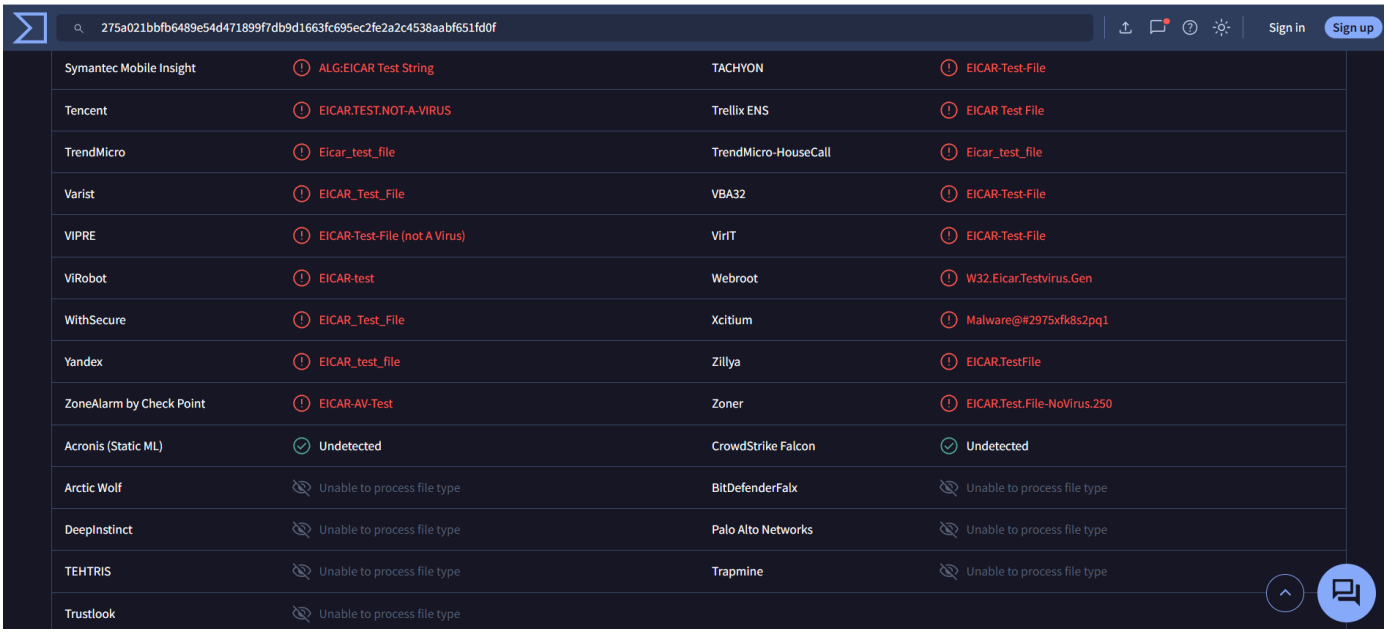


Figure 5. Security vendors' analysis, final section, including engines that did not detect the file.

What this screenshot shows:

- Tencent labels it "EICAR.TEST.NOT-A-VIRUS"; most remaining vendors again name it an EICAR test file.
- Acronis (Static ML) and CrowdStrike Falcon show Undetected. These are the 2 engines that account for 66 / 68 rather than 68 / 68.
- Seven engines, including Palo Alto Networks and DeepInstinct, show "Unable to process file type". They are not counted in the 68, which is why the ratio is 66/68 even though more vendors are listed.
- The lesson: a detection ratio is a count of opinions, not a verdict. Two well-known engines not detecting a file does not make it clean, and 66 detecting it does not make it dangerous.

3.4 File Details and History

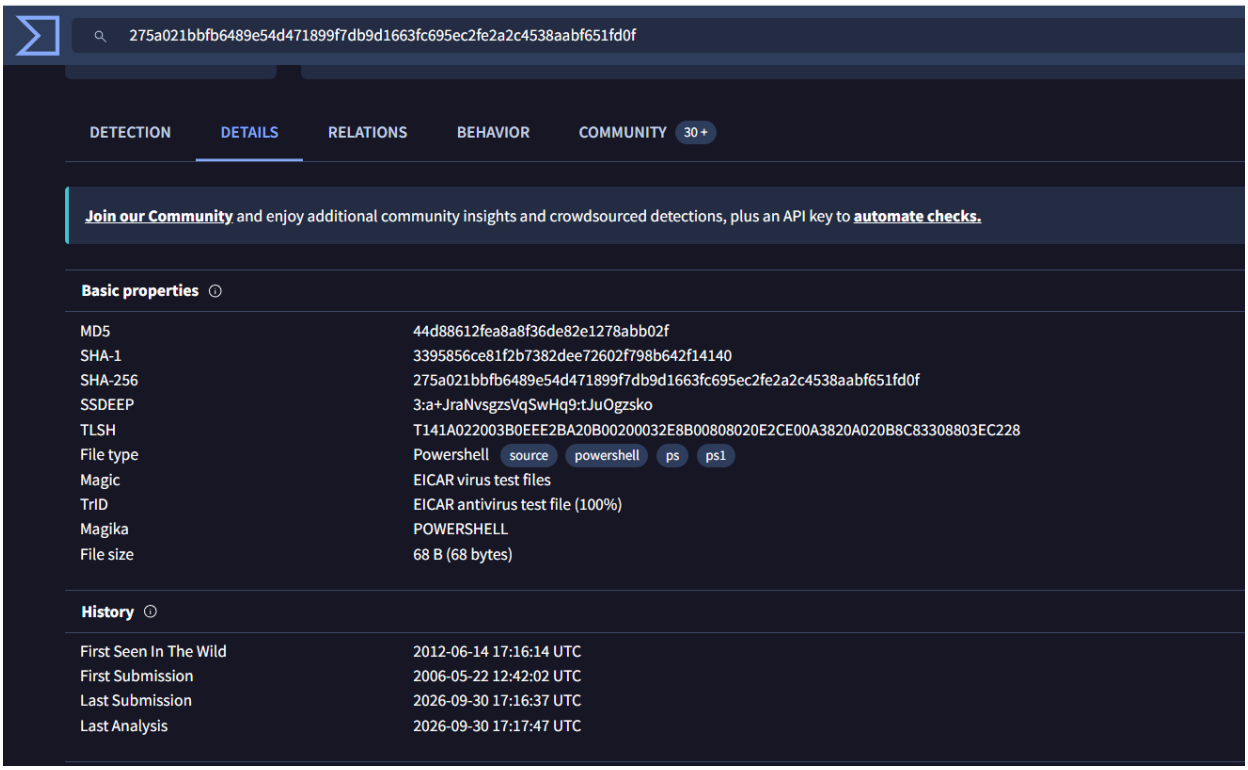


Figure 6. The Details tab: basic properties and submission history.

What this screenshot shows:

- The MD5, SHA-1 and SHA-256 values match the three hashes calculated for this report exactly, confirming this is the correct file record.
- Magic reports "EICAR virus test files" and TrID reports "EICAR antivirus test file (100%)": two independent file-identification tools identify it as the test file.
- File type and Magika report PowerShell. A 68-byte text string can be misread by automated classifiers, which is also the most likely reason for the powershell tag in Figure 2. Where tools disagree, the analyst weighs the specific identifications (Magic, TrID) above the generic one.
- History shows First Submission 2006-05-22 (the earliest time anyone uploaded it to VirusTotal) and First Seen In The Wild 2012-06-14 (the earliest time VirusTotal observed it in real-world use). A file known to VirusTotal for about 20 years is the opposite of a newly emerging threat.
- Last Submission 2026-09-30 17:16:37 UTC and Last Analysis 2026-09-30 17:17:47 UTC (22:46:37 and 22:47:47 IST) show that someone else submitted it minutes before this

lookup, which is expected for a file used worldwide for testing. This lookup was a search only and submitted nothing.

3.5 Community Activity

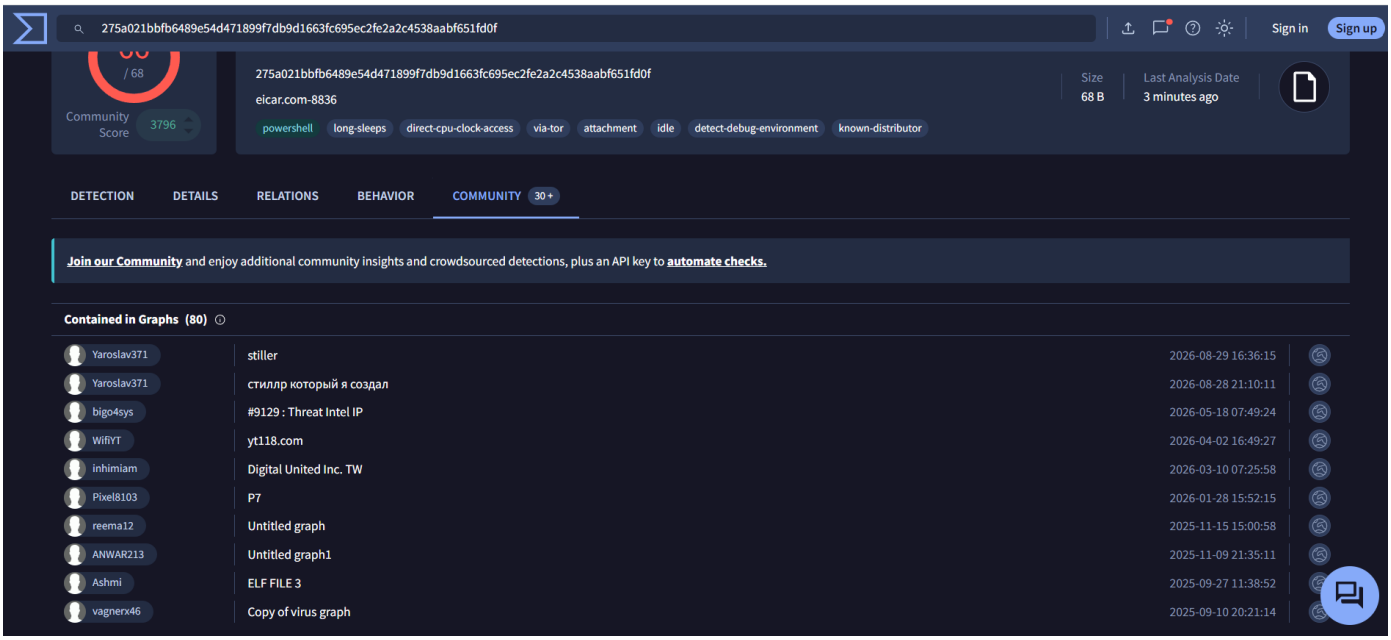


Figure 7. The Community tab: VirusTotal Graphs that include this file.

What this screenshot shows:

- The Community tab shows 30+ items, and the file is "Contained in Graphs (80)". Graphs are investigation diagrams that VirusTotal users build and share.
- Graph titles such as "Untitled graph", "Copy of virus graph" and "ELF FILE 3" are user-created and unverified. EICAR appears in many graphs because people use it for practice and testing.
- Community content can add useful leads, but it is crowd-sourced and must never be treated as confirmation on its own.

4. Source 2: MISP Warninglist

Step 2: Check whether the indicator is known to be non-actionable

MISP (originally the Malware Information Sharing Platform) is an open-source threat-intelligence and sharing platform used by many SOCs and CERTs. Its warninglists are published on GitHub and automatically warn analysts when an indicator is known to be benign or a common false positive.

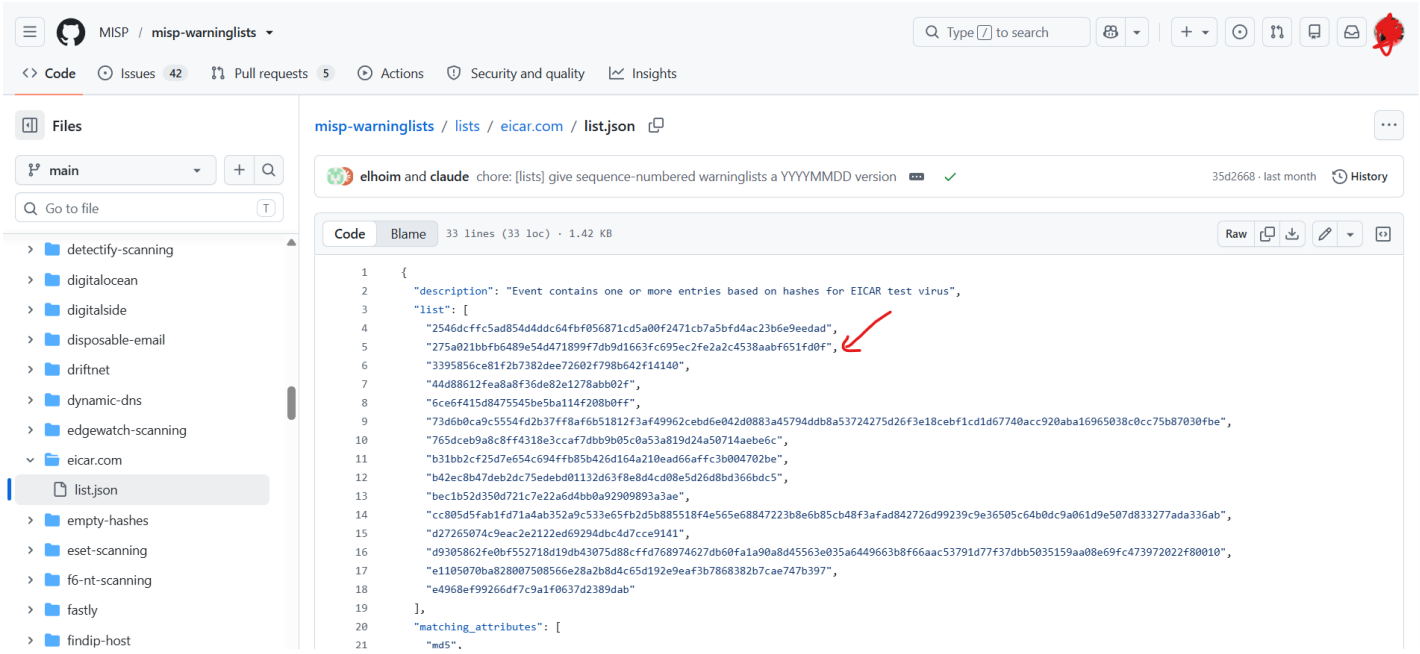


Figure 8. The MISP warninglist for EICAR hashes (misp-warninglists/lists/eicar.com/list.json) on GitHub.

What this screenshot shows:

- The file path misp-warninglists / lists / eicar.com / list.json shows this is the official MISP warninglist for EICAR.
- The description reads "Event contains one or more entries based on hashes for EICAR test virus": when a MISP event contains any of these hashes, analysts are warned that it relates to the EICAR test file.
- Line 5 (marked with the red arrow) is exactly the SHA-256 searched on VirusTotal. Line 6 is the SHA-1 and line 7 the MD5 from Section 2, so all three hashes of this IOC are on the list.
- The latest commit is dated last month, showing the list is actively maintained.

This is the strongest single piece of context in the exercise: an independent, community-maintained source that exists specifically to tell analysts this indicator should not be treated as a real threat.

5. Enrichment Summary

Context	Finding	Effect on priority
Detection ratio	66 / 68 engines	Raises it, if viewed alone
Vendor labels	Mostly "EICAR test" / "not a virus"	Lowers it
File identification	Magic and TrID: EICAR test file; 68 bytes	Lowers it
Distributor	Known distributor (OffSec Services Limited)	Lowers it
Age	First submitted 2006; first seen in the wild 2012	Lowers it: long known, not emerging
MISP warninglist	All three hashes listed as EICAR test virus	Lowers it strongly
Automated tags	powershell, via-tor, long-sleeps and others	Unreliable; do not act on them without corroboration

Default assessment: low priority, benign test indicator.

6. Context That Would Change the Priority

This is the core of the task. The indicator's reputation is fixed, but its priority depends on the environment in which it appears. An analyst would ask the following:

Additional context	If the answer is...	Priority
Was an antivirus or SOC test scheduled on this host?	Yes, and it matches the change record	Low: close as expected test activity
Where did the file appear?	A test VM or the security team's machines	Low
	A production server, or a finance or executive laptop	Raise: why is a test file there?
How did it arrive?	Downloaded by the security team for testing	Low
	Email attachment, unknown download or USB device	Raise: may be probing email or endpoint defenses
How many hosts?	One	Low
	Many hosts at once, with no scheduled test	Raise: could indicate someone mapping which endpoints are protected
Was it blocked?	Yes, antivirus quarantined it	Low: defenses worked as designed
	No, it was not detected	Raise: the endpoint's antivirus may be disabled or misconfigured
Is it inside another file?	No, the plain 68-byte string	Low
	Embedded in an archive or document	Investigate: EICAR is sometimes used to test whether scanners inspect archives

The most important entry is the undetected case. An EICAR alert where antivirus did not block the file is a genuine finding even though the file itself is harmless, because it shows the endpoint's protection is not working. The priority comes from what the indicator reveals about the environment, not from the indicator itself.

7. The Wazuh Side: VirusTotal Integration

Wazuh can perform this enrichment automatically. According to Wazuh's documentation, the integration works like this:

- File Integrity Monitoring (the syscheck module from Day 23) detects a file being added or modified in a monitored directory.
- Wazuh extracts the file's hash from the FIM alert and sends it to the VirusTotal API. Only the hash is sent, not the file.
- VirusTotal's response generates a new Wazuh alert, for example reporting how many engines detected the file, or that the file is not in VirusTotal's database.

Step 3: **Enable the integration on the Wazuh manager (ossec.conf)**

```
<integration>
  <name>virustotal</name>
  <api_key>API_KEY</api_key>
  <group>syscheck</group>
  <alert_format>json</alert_format>
</integration>
```

- name virustotal: selects Wazuh's built-in VirusTotal integration.
- api_key: the analyst's personal VirusTotal API key, available with a free account.
- group syscheck: only FIM alerts trigger a lookup, so every file change in a monitored folder is checked.
- alert_format json: the format in which alerts are passed to the integration.

Wazuh's documentation notes the free public API is limited to 4 requests per minute and 500 per day, and is not permitted for commercial use. In a busy environment this limit is reached quickly, which is why production SOCs use paid threat-intelligence feeds or restrict which directories trigger lookups.

Applied to this indicator: if the EICAR string were saved into a monitored folder on an enrolled agent, Wazuh would raise a FIM alert, send the SHA-256 to VirusTotal, and raise a VirusTotal alert reporting the detections. The analyst would then apply exactly the context in Section 6. Wazuh automates the lookup; it does not supply the judgement.

8. Link to Day 27: The Pyramid of Pain

The Pyramid of Pain, a widely used model by security researcher David Bianco, ranks indicators by how much difficulty blocking them causes an attacker. File hashes sit at the bottom: an attacker can change a hash by altering a single byte. Tactics, techniques and procedures (TTPs), which Day 27 mapped with MITRE ATT&CK, sit at the top because they are hard to change. Hash enrichment is fast and useful for triage, but on its own it is the easiest form of detection for an attacker to evade.

9. Summary

IOC	SHA-256 275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f
Identity	EICAR antivirus test file, 68 bytes
VirusTotal	66 / 68 detections; label virus.eicar/test; first submitted 2006-05-22
MISP	Listed on the EICAR warninglist (SHA-256, SHA-1 and MD5)
Default priority	Low: known benign test file
Priority raisers	Unexpected host, unexpected delivery method, many hosts, not blocked, embedded in other files
Wazuh	VirusTotal integration via FIM; described, not run (no manager deployed)

10. Conclusion

On detection count alone, the EICAR hash looks like serious malware: 66 of 68 engines flagged it. Two independent context sources show it is the harmless antivirus test file, so its default priority is low. The skill this task develops is recognising that reputation data is only the starting point. Priority is set by context: where the indicator appeared, how it arrived, how many hosts are affected, and whether defenses stopped it.

Two lessons carry into real SOC work. First, never upload sensitive files to public services; search by hash instead. Second, treat automated tags, generic machine-learning verdicts and community content as leads to corroborate, not as conclusions.

References

- VirusTotal file report for SHA-256 275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f, viewed 30 September 2026.
- MISP warninglists: lists/eicar.com/list.json, github.com/MISP/misp-warninglists.
- Wazuh documentation: VirusTotal integration (Malware detection).
- EICAR test file: standard test string developed by EICAR (European Institute for Computer Antivirus Research) and CARO (Computer Antivirus Research Organization).
- ThreatDown (Malwarebytes): Why you shouldn't automate your VirusTotal uploads (access to uploaded files).
- Day 23 (File Integrity Monitoring) and Day 27 (MITRE ATT&CK Mapping) reports.